

Full-result packet for arXiv:2603.11040

Exact one-point faithfulness and classification of all optimizers

Status. full_solution_likely_valid. This packet sharpens the one-point thresholding theorem in Sujit Sakharam Damase and James Eldred Pascoe, *On positive definite thresholding of correlation matrices*, arXiv:2603.11040v1.

Source problem. Fix $n \geq 3$, put $\alpha = (n - 2)/2$, and write

$$\mathcal{G}_k(t) = \frac{C_k^{(\alpha)}(t)}{C_k^{(\alpha)}(1)}$$

for the normalized Gegenbauer polynomial, so $\mathcal{G}_k(1) = 1$ and $\mathcal{G}_1(t) = t$. Schoenberg's theorem says that the continuous unital positive definite functions on S^{n-1} are exactly

$$f(t) = \sum_{k=0}^{\infty} a_k \mathcal{G}_k(t), \quad a_k \geq 0, \quad \sum_{k=0}^{\infty} a_k = 1.$$

For $K \subset [-1, 1)$, the source defines the faithfulness constant $\tau_{K,n}$ as the largest possible linear coefficient a_1 among such functions vanishing on K . For $K = \{\varepsilon\}$, with $\varepsilon > 0$ small, the source constructs a degree-two function and proves only

$$\tau_{\{\varepsilon\},n} \geq \frac{|\mathcal{G}_2(\varepsilon)|}{\varepsilon + |\mathcal{G}_2(\varepsilon)|}.$$

The question addressed here is the exact value and the equality case.

Result. The one-point problem is an infinite-dimensional linear program with only one interpolation constraint. Its optimum is determined by the most negative Gegenbauer value at the threshold point. Moreover, every optimizer is supported on degree one and the degrees attaining that minimum. A uniform Darboux estimate then shows that, for every fixed $n \geq 3$, degree two is the unique minimizing degree for all sufficiently small positive thresholds. Thus the source's lower bound is locally an equality, and its quadratic witness is the unique optimizer.

Lemma 1. *For every $n \geq 3$ and $\varepsilon \in (0, 1)$, the number*

$$m_n(\varepsilon) := \min_{k \in \mathbb{Z}_{\geq 0} \setminus \{1\}} \mathcal{G}_k(\varepsilon)$$

is well defined and is strictly negative.

Proof. The existence theorem in arXiv:2603.11040 supplies a nonzero continuous positive definite function vanishing at ε . After normalizing at 1, its Schoenberg coefficients form a probability vector. Since $\mathcal{G}_0(\varepsilon) = 1$ and $\mathcal{G}_1(\varepsilon) = \varepsilon > 0$, the identity $\sum_k a_k \mathcal{G}_k(\varepsilon) = 0$ forces $\mathcal{G}_j(\varepsilon) < 0$ for some $j \neq 1$.

For fixed $\varepsilon \in (-1, 1)$, the classical Darboux estimate for normalized Gegenbauer polynomials gives $\mathcal{G}_k(\varepsilon) \rightarrow 0$ as $k \rightarrow \infty$ when $n \geq 3$. Because at least one term is negative, the infimum is negative and is attained among finitely many indices. It is therefore the displayed minimum. $\square$

Theorem 2 (Exact one-point faithfulness). *Let $n \geq 3$ and $0 < \varepsilon < 1$. Then*

$$\tau_{\{\varepsilon\},n} = \frac{-m_n(\varepsilon)}{\varepsilon - m_n(\varepsilon)}.$$

Let $M_n(\varepsilon) = \{k \neq 1 : \mathcal{G}_k(\varepsilon) = m_n(\varepsilon)\}$. A Schoenberg coefficient vector $a = (a_k)$ is optimal if and only if

$$a_1 = \frac{-m_n(\varepsilon)}{\varepsilon - m_n(\varepsilon)}, \quad \sum_{k \in M_n(\varepsilon)} a_k = \frac{\varepsilon}{\varepsilon - m_n(\varepsilon)},$$

and every other coefficient vanishes. In particular, an optimizer can always be chosen to use exactly two Gegenbauer modes.

Proof. Let $f = \sum a_k \mathcal{G}_k$ be feasible. Using $f(\varepsilon) = 0$, the definition of $m = m_n(\varepsilon)$, and $\sum_k a_k = 1$, we obtain

$$0 = a_1 \varepsilon + \sum_{k \neq 1} a_k \mathcal{G}_k(\varepsilon) \geq a_1 \varepsilon + m \sum_{k \neq 1} a_k = a_1 \varepsilon + m(1 - a_1).$$

As $m < 0$, this rearranges to

$$a_1 \leq \frac{-m}{\varepsilon - m}.$$

Choose $j \in M_n(\varepsilon)$ and set

$$a_1 = \frac{-m}{\varepsilon - m}, \quad a_j = \frac{\varepsilon}{\varepsilon - m}, \quad a_k = 0 \quad (k \notin \{1, j\}).$$

These coefficients are nonnegative, sum to one, and satisfy $a_1 \varepsilon + a_j m = 0$. They therefore define a feasible continuous positive definite function attaining the upper bound.

Finally, equality in the displayed estimate is possible precisely when every positive coefficient outside degree one is carried by an index for which $\mathcal{G}_k(\varepsilon) = m$. This is exactly the asserted classification. $\square$

Theorem 3 (Local sharpness and uniqueness of the quadratic witness). *For every fixed $n \geq 3$, there is $\delta_n > 0$ such that, whenever $0 < \varepsilon < \delta_n$,*

$$m_n(\varepsilon) = \mathcal{G}_2(\varepsilon) = \frac{n\varepsilon^2 - 1}{n - 1},$$

and degree two is the unique minimizing degree. Consequently

$$\tau_{\{\varepsilon\},n} = \frac{1 - n\varepsilon^2}{1 - n\varepsilon^2 + (n - 1)\varepsilon}$$

for all such ε , and the unique optimal function is

$$f_\varepsilon(t) = \tau_{\{\varepsilon\},n} t + (1 - \tau_{\{\varepsilon\},n}) \frac{nt^2 - 1}{n - 1}.$$

Proof. At zero, parity gives $\mathcal{G}_{2r+1}(0) = 0$, while the duplication formula for the Gamma function gives

$$\mathcal{G}_{2r}(0) = (-1)^r \frac{(1/2)_r}{((n-1)/2)_r} \quad (r \geq 0).$$

For $n \geq 3$, the absolute values on the right strictly decrease with r . Hence

$$\mathcal{G}_2(0) = -\frac{1}{n-1} < \mathcal{G}_k(0) \quad (k \neq 1, 2),$$

with a positive gap: the even subsequence decreases in absolute value and the odd values are zero.

The Darboux estimate is uniform for t in a compact subinterval of $(-1, 1)$, so $\mathcal{G}_k(t) \rightarrow 0$ uniformly for t in a fixed neighborhood of zero. Choose the tail large enough that no tail polynomial can close the gap to $\mathcal{G}_2(0)$. The remaining family is finite, and continuity preserves the strict inequalities after shrinking the neighborhood. Thus there is $\delta_n > 0$ for which degree two remains the unique minimizer.

The formula $\mathcal{G}_2(t) = (nt^2 - 1)/(n - 1)$, followed by Theorem 2, gives the displayed value and the unique equality case. $\square$

Checks and scope. The proof uses exactly the source’s continuous Schoenberg class and the same normalization of the faithfulness coefficient. It is restricted to $n \geq 3$, where normalized Gegenbauer values at an interior point tend to zero; the circle $n = 2$ has a distinct Chebyshev/Diophantine endpoint and is not claimed here. The result sharpens the source’s one-point theorem but does not resolve its final conjecture about discontinuous positive definite functions.

Novelty check. Before promotion, the run indexes were searched for arXiv:2603.11040, the phrases “faithfulness constant” and “one-point thresholding,” and the Gegenbauer/positive-definite combinations; no earlier packet was found. A bounded primary-source search on 2026-07-19 found only arXiv:2603.11040v1 and its companion arXiv:2602.09010. The current source record still describes “bounds” at single points, and v1 contains only the quadratic lower bound, not the exact formula or equality classification above.

Human-review recommendation. Review as a short exact extension of the source’s one-point theorem. The linear-program argument is elementary; the only analytic point to audit is the standard uniform-on-compacts Darboux estimate used to make degree two uniformly optimal near zero.

References

- [1] S. S. Damase and J. E. Pascoe, *On positive definite thresholding of correlation matrices*, arXiv:2603.11040v1, 2026.
- [2] S. S. Damase and J. E. Pascoe, *Complete discrete Schoenberg–Delsarte theory for homogeneous spaces*, arXiv:2602.09010v1, 2026.
- [3] I. J. Schoenberg, *Positive definite functions on spheres*, Duke Math. J. **9** (1942), 96–108.